

6 Magnet Rampdown to Zero Field

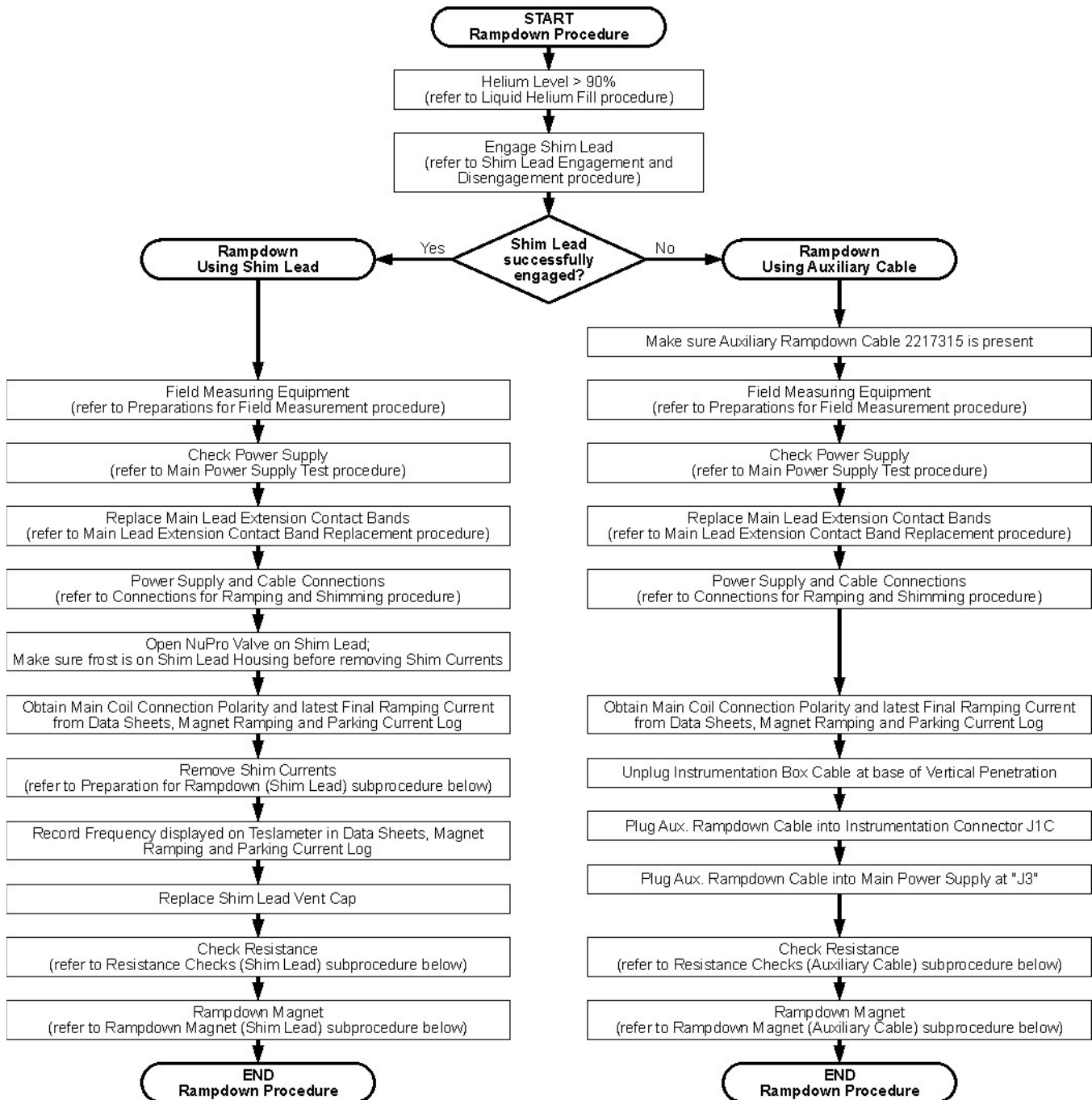
6.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	300 mins	Not Applicable

6.2 Overview

Rampdown is performed using the Shim Lead unless the Shim Lead cannot be engaged into the Shim Lead Connector (P100) in the magnet Vertical Penetration due to ice or damage to the P100's Sav-Con Connector, in which case rampdown is performed using Auxiliary Rampdown Cable 2217315. See Illustration 5-14.

Illustration 5-14: Rampdown Flowchart



Procedures Referenced:	
Rampdown Using Shim Lead	Rampdown Using Auxiliary Cable
Chapter 3, Section 3.2, Liquid Helium Fill	Chapter 3, Section 3.2, Liquid Helium Fill
Chapter 3, Section 5, Shim Lead Engagement and Disengagement	Chapter 3, Section 5, Shim Lead Engagement and Disengagement
Chapter 3, Section 4, Preparations for Field Measurement	Chapter 3, Section 4, Preparations for Field Measurement

Procedures Referenced:	
Rampdown Using Shim Lead	Rampdown Using Auxiliary Cable
Main Power Supply Tests: Chapter 4, Section 8.2, TCR 7.5T750 Main Power Supply Tests, for 750 amp power supplies or Chapter 4, Section 8.3, ESS7.5-1000-2-D-1236 Main Power Supply Tests,	Main Power Supply Tests: Chapter 4, Section 8.2, TCR 7.5T750 Main Power Supply Tests, for 750 amp power supplies or Chapter 4, Section 8.3, ESS7.5-1000-2-D-1236 Main Power Supply Tests, for 1000 amp power supplies
Section 4, Main Lead Extension Contact Band Replacement	Section 4, Main Lead Extension Contact Band Replacement
Chapter 3, Section 6, Connections for Ramping and Shimming	Chapter 3, Section 6, Connections for Ramping and Shimming
Chapter 8, Section 5, Magnet Ramping and Parking Current Log	Chapter 8, Section 5, Magnet Ramping and Parking Current Log
Section 6.4.1.1, Preparation for Rampdown (Shim Lead)	Section 6.4.2.1, Preparation for Rampdown (Auxiliary Cable)
Section 6.4.1.2, Resistance Checks (Shim Lead Rampdown)	Section 6.4.2.2, Resistance Checks (Auxiliary Cable)
Section 6.4.1.3, Magnet Rampdown (Shim Lead)	Section 6.4.2.3, Magnet Rampdown (Auxiliary Cable)

6.3 Preliminary Requirements

6.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Nonabsorbent protective clothing (long sleeve shirt and pants)	1 set per person	-	-	-
Nonferrous Shoes	1 pair per person	-	-	-
750 Amp TCR 7.5T750 Main Power Supply (46-260776G3) or 1000 Amp ESS 7.5-1000-2-D-1236 Main Power Supply (46-260776G4)	1	-	-	-
750 Amp Ramp Cart/Cable Kit (2135435) or 1000 Amp Ramp Cart/Cable Kit w/ Ground (2353394)	1	-	-	-
Fixed Main Lead Extension (46-294204G1) or Flexible Main Lead Extension (2287029) (for low ceiling height sites)	2	-	-	-
Hold-Down Tool (2193922) or Hold-Down Tool, Short (2276755) (for low ceiling height sites)	1	-	-	-

Item	Qty	Effectivity	Part#	Manufacturer
Retractable Shim Lead Assembly (2173354) or Retractable Shim Lead Assembly, Short (2285073) (for low ceiling height sites)	1	-	-	-
Digital Voltmeter (DVM)	1	-	-	-
Cardboard box or nonferromagnetic frame		-	-	-
Warning Sign & Label Kit	1	-	46-258770G4	-
Portable Oxygen Monitor	1	-	2287000	-
Cryogen Safety Wear Kit	1 kit per person	-	46-271137G1	-
Field Shim Case & Cable Kit (if not already ordered with the magnet)	1	-	2135558	-
Auxiliary Rampdown Cable	1	-	2217315	-
MetroLab Teslameter Kit	1	-	46-251865G4	-
#5 MetroLab Probe (included in MetroLab Teslameter Kit 46-251865G4)	1	-	46-255839P104	-
Nonmagnetic Tools Kit	AR	-	46-320273G4	-

6.3.2 Consumables

No consumables required.

6.3.3 Replacement Parts

No replacement parts required.

6.3.4 Safety



⚠ DANGER

POTENTIAL FATAL INJURY!

MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, BEFORE STARTING THIS PROCEDURE.


! DANGER

POTENTIAL FATAL INJURY!

MAKE SURE TO FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, SECTION 2.4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS, BEFORE AND WHILE PERFORMING THIS PROCEDURE.


! DANGER

POTENTIAL FATAL INJURY!

HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, SECTION 2.5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION, BEFORE STARTING THIS PROCEDURE.


! WARNING

ASPHYXIATION HAZARD!

THIS PROCEDURE GENERATES ODORLESS, COLORLESS HELIUM GAS THAT DISPLACES OXYGEN IN THE AIR. MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS ON — OR THE HATCH IS OPENED IF A MOBILE VAN — BEFORE STARTING THIS PROCEDURE.


! WARNING

POTENTIAL PROJECTILE HAZARD

FERROMAGNETIC OBJECTS IN A STRONG MAGNETIC FIELD CAN BECOME DANGEROUS PROJECTILES.

REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. MAKE SURE ALL EQUIPMENT AND TOOLS USED IN THE MAGNET ROOM ARE NON-FERROMAGNETIC. LOCATE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND ROOM GEOMETRY ALLOWS. DO NOT ALLOW ANYONE CARRYING FERROMAGNETIC MATERIALS — INCLUDING CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES — INSIDE THE 5 GAUSS ZONE.


! WARNING

EXPLOSION HAZARD!

LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN, PRODUCING A HIGHLY COMBUSTIBLE OXYGEN LIQUID. NO SMOKING IN MAGNET AND CRYOGEN STORAGE ROOMS!


! WARNING

POTENTIAL QUENCH HAZARD

THE MAGNET MAY QUENCH, RAPIDLY RELEASING CRYOGENIC GASES, IF ITS LIQUID HELIUM LEVEL DROPS TOO LOW DURING THIS PROCEDURE. MAKE SURE THE MAGNET $\geq$ 90% FULL OF LIQUID HELIUM BEFORE STARTING THIS PROCEDURE.



⚠ WARNING

POTENTIAL COLD BURN HAZARD
EXPOSURE TO CRYOGENS AND CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE.
WEAR PROTECTIVE CLOTHING, NONABSORBENT GLOVES AND GOGGLES WHEN PERFORMING THIS PROCEDURE.



⚠ WARNING

ELECTRICAL SHOCK HAZARD!
TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER. MAKE SURE INPUT POWER TO THE MAIN POWER SUPPLY IS DISCONNECTED WHEN CONNECTING THE MAIN LEAD EXTENSIONS.



NOTICE

Locate the Main Power Supply OUTSIDE the Magnet Room.

6.3.5 Required Conditions

Condition	Reference	Effectivity
Make sure the area is secure and that ALL required Warning Signs are posted before starting this procedure.	-	-
The removal of some magnet enclosure components may be required to complete this procedure. Refer to Section 5, Upper and Side Enclosure Removal, before continuing with this procedure if cover removal is required.	-	-
Make sure the helium level in the magnet is $\geq 90\%$ before starting this procedure.	-	-
Locate the Main Power Supply outside the magnet room and remove all loose ferromagnetic materials from the magnet room before starting this procedure.	-	-

6.4 Procedure

6.4.1 Magnet Rampdown Using Shim Lead

6.4.1.1 Preparation for Rampdown (Shim Lead)

1. "Top Fill" the magnet to $\geq 90\%$ helium level in conformance with Chapter 3, Section 3.2, Liquid Helium Fill, before starting this procedure.
2. Make sure the Shim Lead Assembly is "Engaged" in conformance with Chapter 3, Section 5, Shim Lead Engagement and Disengagement.

NOTE: If the Shim Lead cannot be engaged, ramp down the magnet in conformance with Section 6.4.2.

3. Set up MetroLab Teslameter 46-255839P2 and #5 Teslameter Probe 46-255839P104 in conformance with Chapter 3, Section 4, Preparations for Field Measurement. The probe can be mounted to either a cardboard box or nonferromagnetic frame since only the overall field strength is monitored during ramping.
4. Check the Main Power Supply in conformance with the Main Power Supply Tests section (Chapter 4, Section 8.2, TCR 7.5T750 Main Power Supply Test, for 750 amp power supplies or Chapter 4, Section 8.3, ESS7.5-1000-2-D-1236 Main Power Supply Tests, for 1000 amp power supplies). Verify that the Switch Heater currents are set at $810 \text{ mA} \pm 10 \text{ mA}$.
5. Make sure the input power cables for the power supplies are disconnected.
6. Replace the Main Lead Contact Bands in conformance to Section 4, Main Lead Extension Contact Band Replacement.
7. Connect the Shim Coil Power Supply to the magnet by making all cable connections in conformance with Chapter 3, Section 6, Connections for Ramping and Shimming.
8. Open the Shim Lead's Nupro Vent Valve and allow frost to appear on the Shim Lead Connector Housing before removing shim currents.



NOTICE

Do not connect input power to the Superconducting Shim Coil Power Supply until it is verified that ALL current controls are set to zero (turned fully counterclockwise) and the supply is turned off.

9. Connect the input power supply cable to the Shim Coil Power Supply.

NOTE: This section allows one to remove all shim currents and find the actual main field to document and use as a target parking frequency when reramping the magnet. This will enable parking the magnet at a more accurate frequency and prevent retuning the rest of the system.



⚠ WARNING

POTENTIAL QUENCH HAZARD

THE MAGNET MAY QUENCH, RAPIDLY RELEASING CRYOGENIC GASES, IF ITS LIQUID HELIUM LEVEL DROPS TOO LOW DURING THIS PROCEDURE.

MAKE SURE THE MAGNET $\geq 90\%$ FULL OF LIQUID HELIUM BEFORE STARTING THIS PROCEDURE.



⚠ WARNING

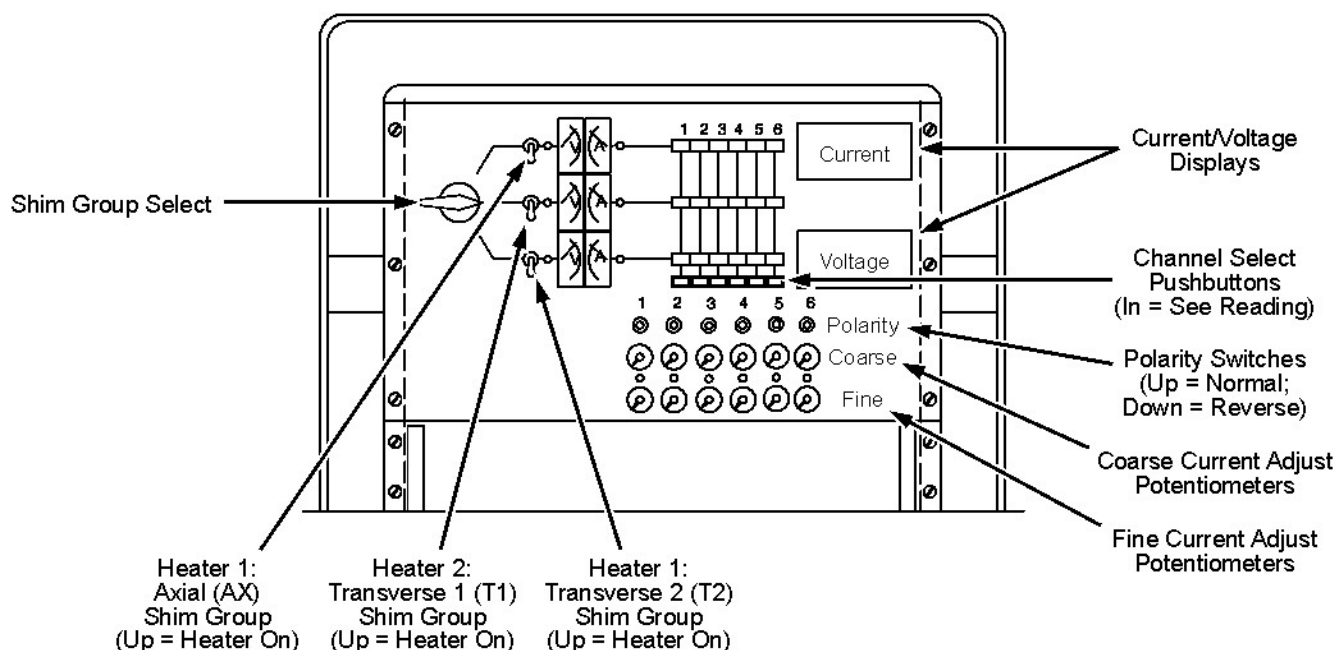
POTENTIAL QUENCH HAZARD

APPLYING POWER THROUGH THE SHIM LEAD BEFORE THE SHIM LEAD IS SUFFICIENTLY COOLED OR WHILE THE SHIM LEAD'S VENT VALVE IS CLOSED MAY RESULT IN A MAGNET QUENCH.

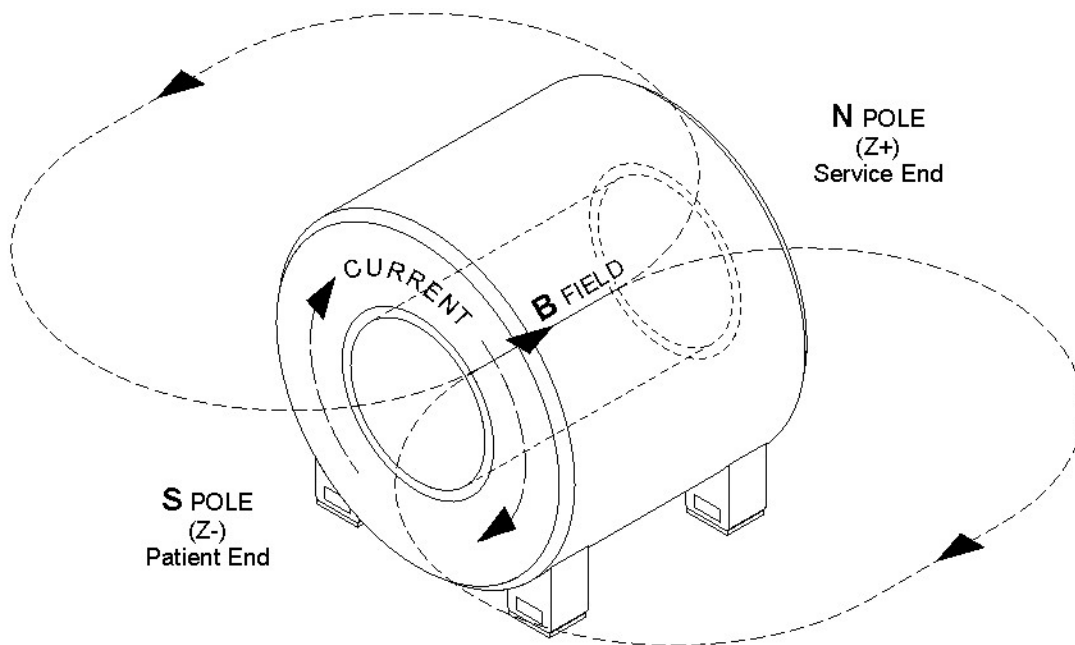
MAKE SURE THE SHIM LEAD'S NUPRO VENT VALVE IS OPEN AND FROSTING IS VISIBLE ON THE SHIM LEAD HOUSING BEFORE TURNING ON THE SHIM COIL POWER SUPPLY.

10. Set the Main Power Switch on the Shim Coil Power Supply to the "ON" position.
11. Set the Shim Coil Power Supply (Illustration 5-15) to all last recorded Transverse 1 Currents from Chapter 8, Section 5, Magnet Ramping and Parking Current Log. Make sure the current polarities are correct.

Illustration 5-15: Shim Coil Power Supply Controls (Front Panel)



NOTE: All Main Coil driving voltages provided in this procedure will be equal in magnitude and opposite in polarity for magnets ramped with reversed polarity. Normal polarity is shown in Illustration 5-16.

Illustration 5-16: Magnet Polarity Ramped Normal

12. Set the Transverse 1 Heater Switch on the Shim Power Supply to "I" (on).
13. Verify that the heater currents are set to Transverse 1 = 610 ± 5 mA, Transverse 2 = 610 ± 5 mA and Axial = 710 ± 10 mA. If any current level is not correct, adjust the adjustment screw on the rear of the Shim Power Supply. Allow a few minutes for the heater to drive the switches resistive.
14. Slowly adjust all Shim "Current" controls to zero.

NOTE: The field strength reading on the Teslameter should change as the Axial Shim Currents are adjusted down to zero. If no change in the Teslameter reading is noticed, the Axial Heater Switch may still be "persistent." If this occurs, wait an additional 2 minutes before continuing to decrease currents.

15. Set the Heater Switch to "O" (off).
16. Repeat Step 11 through Step 15 for Transverse 2 and then the Axial Shim Coils.
17. Set the Main Power Switch on the Shim Power Supply to "OFF" and disconnect all cables.

NOTE: The frequency displayed on the Teslameter after the shim currents are removed is the target frequency for when the magnet is reramped.

18. Record the frequency displayed on the Teslameter in Chapter 8, Section 5, Magnet Ramping and Parking Current Log.
19. Close the Shim Lead's Nupro Vent Valve.

6.4.1.2 Resistance Checks (Shim Lead Rampdown)

1. Make sure the input power cable for the Main Power Supply is disconnected.

2. Connect the Main Power Supply and Main Lead Extensions to the magnet by making all cable connections in conformance with Chapter 3, Section 6, Connections for Ramping and Shimming.
3. Set the Heater 1 Main and Heater 2 Shim Axial switches to the "O" (off) position. See Illustration 5-17 (750 amp Main Power Supply) or Illustration 5-18 (1000 amp Main Power Supply).

Illustration 5-17: 750 Amp TCR 7.5T750 Superconducting Main Coil Service Power Supply Cabinet (46-260776G3)

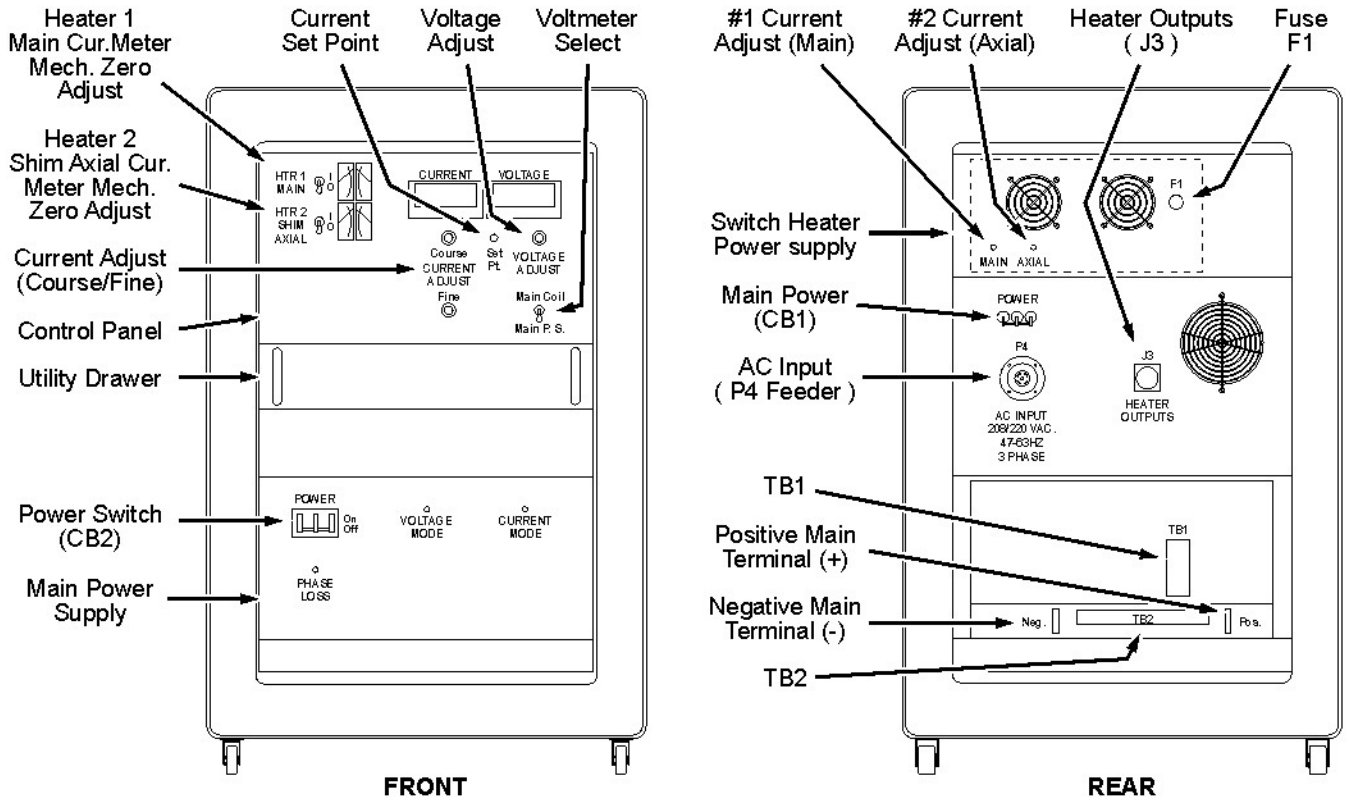
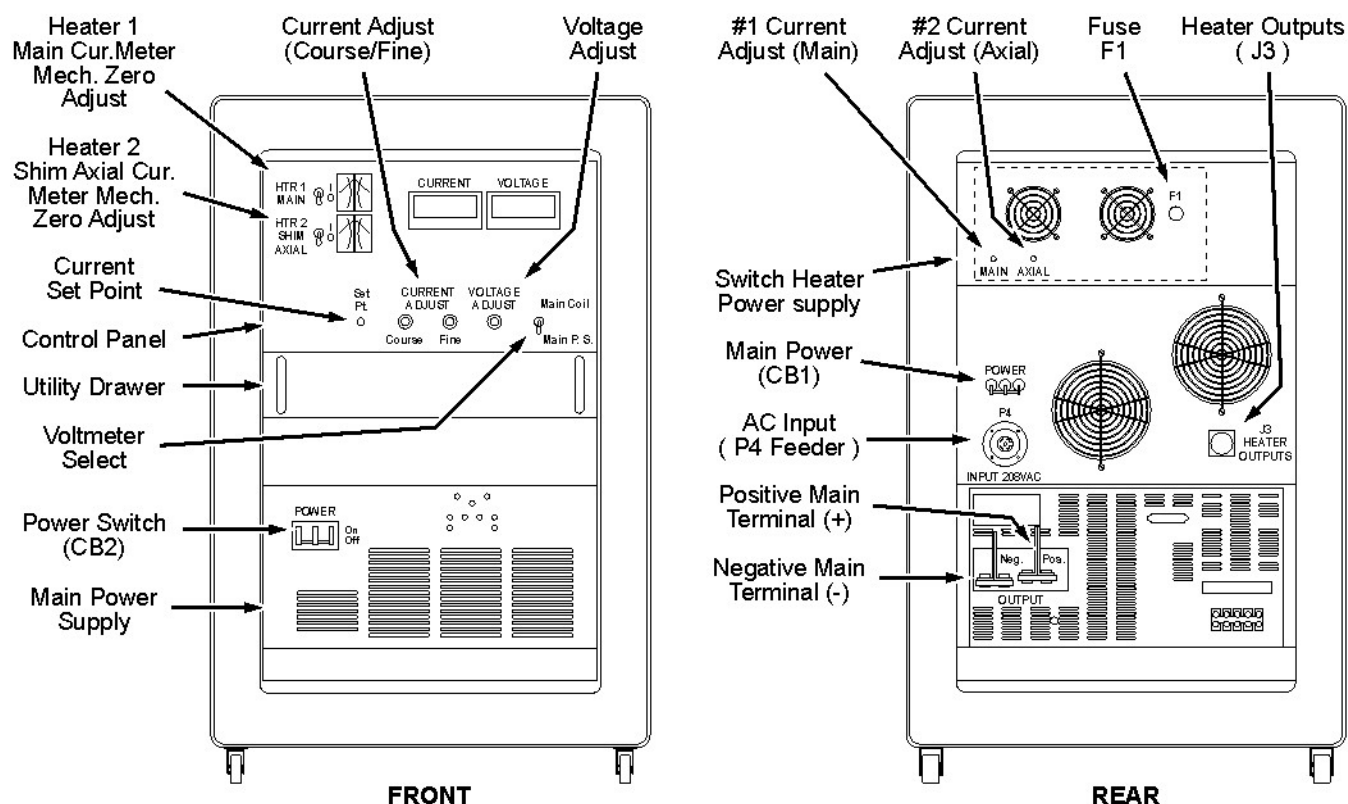


Illustration 5-18: ESS7.5-1000-2-D-1236 Superconducting Main Coil Service Power Supply Cabinet (46-260776G4)



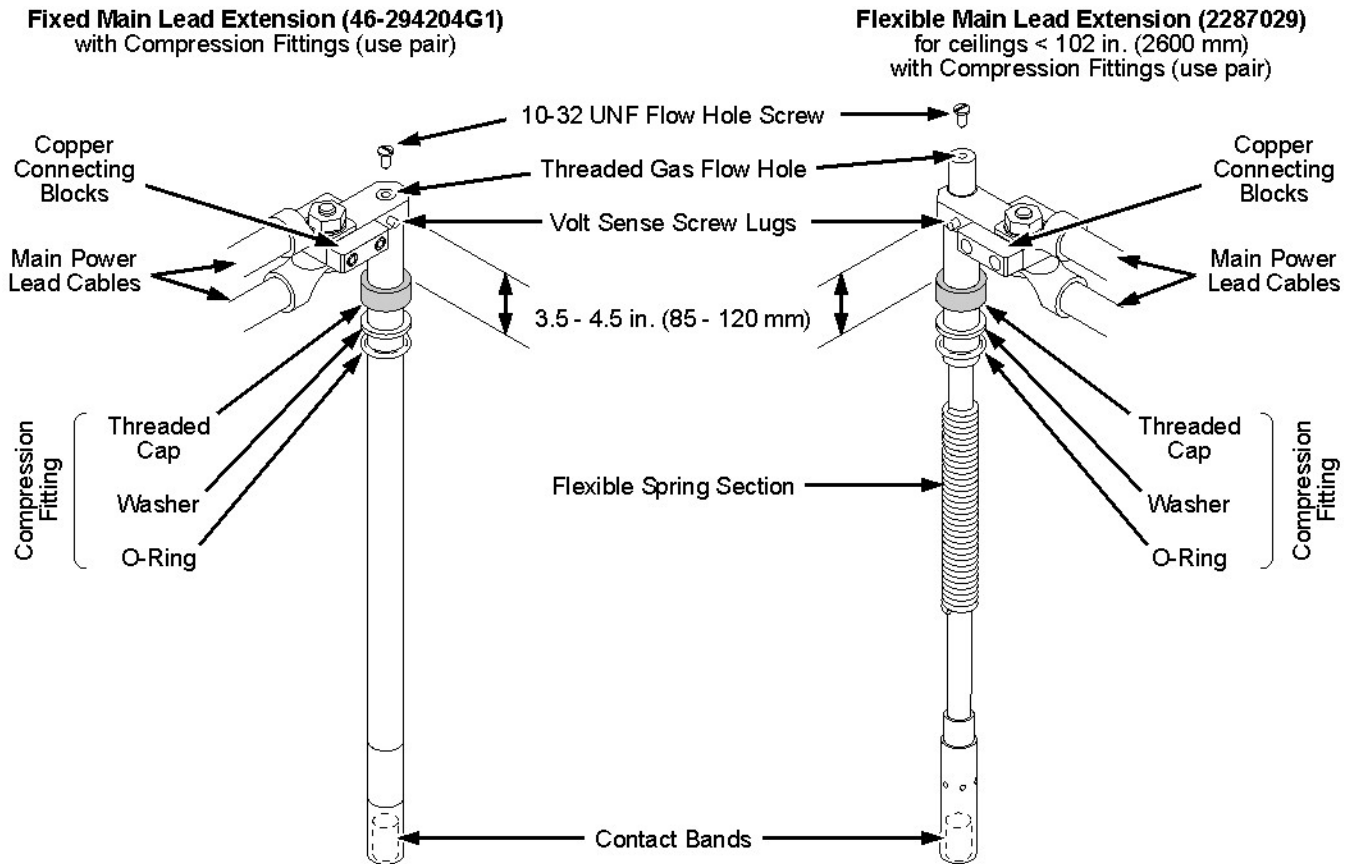
4. Set the Current Adjust and Voltage Adjust controls to "O" (full CCW).
5. Connect the input power cable for the Magnet Power Supply.



WARNING

POTENTIAL MAGNET QUENCH
TURNING ON THE MAIN HEATER SWITCH WHILE PERFORMING THESE
RESISTANCE CHECKS WILL CAUSE A MAGNET QUENCH.
MAKE SURE THE MAIN HEATER SWITCH STAYS OFF DURING RESISTANCE
CHECKS.

6. Set the Main Power and Power On switches to "ON."
7. Set the Heater 2 Shim Axial Switch to "I" (on) and observe current rise in an analog ammeter (810 ± 10 mA) to verify circuit continuity. Make sure the Main Heater Switch is off.
8. Connect a digital voltmeter (DVM) to the end of the Voltage Sense Leads (identified in Illustration 5-19).

Illustration 5-19: Main Lead Extensions

9. Set the Current Adjust Controls on the Main Power Supply to maximum (full CW).
10. Observe the Digital Current Meter and slowly turn the Voltage Control (CW) to set a 500 A current through the Main Leads, Main Lead Extensions and the persistent Main Switch.
11. Record the voltage reading on the DVM in Chapter 8, Section 5, Magnet Ramping and Parking Current Log.



CAUTION

Potential Magnet Quench

Voltage readings > 150 millivolts at 500 amps indicate unacceptable internal contact resistance at the Main Lead Extensions. Higher resistances will add more heat to the magnet, increasing boiloff and possibly causing a quench during ramping. Make sure the voltage readings are ≤ 150 mV at 500 amps.

12. If the DVM reads > 150mV, perform one or more of the following steps to reduce the contact resistance.
 - Wait approximately 1 minute with current running, readings may drop as the Main Lead Extensions cool.
 - Tighten the nuts (one flat at a time) on top of the Hold-Down Tool while measuring DVM readings.

If the reading still exceeds 150 mV:

1. Turn the Voltage and Current Adjust controls to zero (full CCW).
2. Turn off the Main Power Supply input power.
3. Tighten the bolts securing the Ramp Cables to the power supply and Main Leads Extensions.
4. Lift and reseal the Main Leads.

Repeatedly failing the contact resistance check may indicate one of the following:

- The Main Lead Extension Contact Bands replaced as directed in Chapter 3, Section 6, Connections for Ramping and Shimming, are not performing correctly and need replacing in conformance with Section 4, Main Lead Extension Contact Band Replacement.
- The Main Leads are damaged and need replacing.

If changes are made in Step 12, repeat Step 1 through Step 12.

13. Continue to Step 14 after the DVM voltage is < 150 mV.
14. Gradually increase the Voltage Adjust Control to pass 735 A through the Main Leads, Main Lead Extensions and the persistent Main Switch while observing the power supply's Voltmeter. If the voltage exceeds 2.2 V, discontinue the test and perform Step 12 above.
15. Set the power supply's Voltmeter Select Switch to the "MAIN POWER SUPPLY" position. (This will display the output at the output lugs of the power supply monitored.) A voltage < 2.2 V at 735 A indicates acceptable system resistance. If the voltage exceeds 2.2 V during the test, perform Step 12 to adjust contact resistance.
16. Turn the Current Adjust and Voltage Adjust controls off (full CCW) and continue with Section 6.4.1.3.

6.4.1.3 Magnet Rampdown (Shim Lead)



WARNING

POTENTIAL QUENCH HAZARD
INCORRECT CONNECTION POLARITY AND POWER SUPPLY CURRENT SETTINGS CAN CAUSE THE MAGNET'S MAIN COILS TO QUENCH AND CAN BURN UP THE POWER SUPPLY WHEN TURNING ON THE MAIN SWITCH.
MAKE SURE THE CONNECTION POLARITY AND POWER SUPPLY CURRENT SETTINGS ARE THE SAME AS LAST RECORDED IN THE "MAGNET RAMPING AND PARKING CURRENT LOG".



WARNING

POTENTIAL QUENCH HAZARD

IF THE AXIAL SHIM HEATER SWITCH IS NOT “ON” DURING THE ENTIRE MAIN FIELD ADJUSTMENT PROCESS, A MAGNET QUENCH CAN OCCUR AND IRREPARABLE DAMAGE SHIM COIL DAMAGE CAN BE DONE. THE POWER SUPPLY WILL NOT PASS CURRENT TO THE MAIN LEAD EXTENSIONS WITH THE AXIAL SHIM HEATER OFF.

MAKE SURE THE AXIAL SHIM HEATER SWITCH REMAINS “ON” DURING THIS ENTIRE PROCEDURE.



WARNING

POTENTIAL QUENCH HAZARD

THE MAGNET MAY QUENCH IF THE MAIN POWER SUPPLY EXPERIENCES LARGE OUTPUT FLUCTUATIONS AND/OR EXCESSIVE RIPPLE.

MAKE SURE THE MAIN POWER SUPPLY IS ROUTINELY CALIBRATED AT AN APPROVED FACILITY.



NOTICE

If a Quench occurs during change of magnetic field, immediately turn the Voltage Adjust and Current Adjust controls to zero.



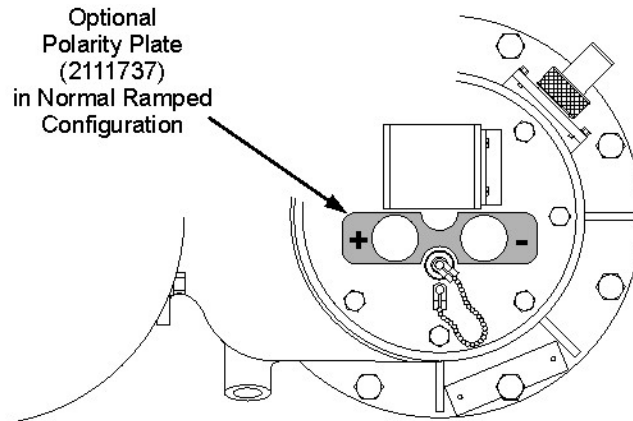
NOTICE

Make sure the polarity recorded Chapter 8, Section 5, Magnet Ramping and Parking Current Log, matches the configuration of the Ramp Polarity Plate (Illustration 5-20). The "Normal" ramp polarity is “+” (positive) on the left and “-” (negative) on the right as viewed from the Coldhead side of the magnet. “Reversed” ramp polarity has “+ on the right and “-” on the left as viewed from the Coldhead side of the magnet. Call the National Support Center if the two do not match.

NOTE: All main coil driving voltages provided in this procedure will be equal in magnitude but opposite in polarity for “reverse ramped” magnets. Normal polarity is shown in Illustration 5-16.

NOTE: Ice will build up around the Ramp Lead Hold-Down Tool flow holes during the rampdown procedure. Remove ice as needed to maintain helium gas flow through the flow holes.

1. Retrieve the Main Coil connection polarity and latest final parking current from Chapter 8, Section 5, Magnet Ramping and Parking Current Log.

Illustration 5-20: Polarity Plate 2111737

2. Make sure the Axial Shim Heater is on and the Main Leads are connected with the polarity indicated in Step 1.

NOTE: The Axial Shim Heater must remain on throughout the rampdown procedure.

3. Set the power supply Voltmeter Select Switch to "MAIN COIL" position.
4. Set the Current Adjust Controls on power supply to:
 - **for 750 amp power supply 46-260776G3:** maximum (full CW).
 - **for 1,000 amp power supply 46-260776G4:** 10 amps more than the parking current retrieved in Step 1.

NOTE: Power supply controls are shown in Illustration 5-17 (750 amp Main Power Supply) and Illustration 5-18 (1000 amp Main Power Supply).

5. Set the Voltage Control to adjust the power supply output current to the parking current value retrieved in Step 1 above.
6. Set the Heater 1 Main Switch to "I" (on).
7. Allow approximately 1 minute for the Main Switch to go normal.
8. When the Main Switch is normal, **slowly** decrease the Voltage Control (CCW) until -0.3 V is observed across the power supply voltmeter ("Main Coil" position).
9. Continue to adjust power supply voltage using the Voltage Control during rampdown in conformance with the voltages and main coil current ranges shown in Table 5-4.

Table 5-4: Rampdown (Using Shim Lead) Profile

Main Coil Voltage	Main Coil Current Range
-0.30 Volts	Start - 720 Amps
-0.50 Volts	720 - 700 Amps
-0.60 Volts	700 - 675 Amps

Table 5-4: Rampdown (Using Shim Lead) Profile (cont'd)

Main Coil Voltage	Main Coil Current Range
-0.75 Volts	675 - 650 Amps
-1.00 Volts	650 - 625 Amps
-1.50 Volts*	625 - 500 Amps
Gradually turn the Voltage Control to zero (full CCW) then turn the Current Controls to zero (full CCW)	500 - 0 Amps
* -1.50 volts is the approximate maximum voltage obtainable during rampdown.	

10. Observe the voltmeter reading with the power supply Voltmeter Select Switch in the "MAIN COIL" position. A zero reading on the power supply digital voltage and current meters indicates that the magnet is fully discharged.
11. When the magnet is full discharged, set the Heater 1 Main and the Heater 2 Shim Axial switches to "O" (off).
12. Set the Main Power and Power On switches to "OFF."
13. Disconnect the input power cables from the Main Power Supply.

NOTE: The Pressure Controller will maintain helium vessel pressure.



! WARNING

POTENTIAL ASPHYXIATION AND COLD BURN HAZARD!
IMPROPER ENGAGEMENT/DISENGAGEMENT OF THE MAIN LEAD EXTENSIONS WITH THE MAIN LEAD PINS CAN CAUSE A RAMPED MAGNET TO QUENCH. INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND, IF THE MAGNET IS AT FIELD, ALLOW THEM TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE THEY CONTACT THE MAIN LEAD PINS.



! WARNING

ELECTRICAL SHOCK HAZARD!
TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.



! WARNING

ELECTRICAL SHOCK HAZARD!
DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.



WARNING

POTENTIAL COLD BURN HAZARD
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE.
WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING
INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

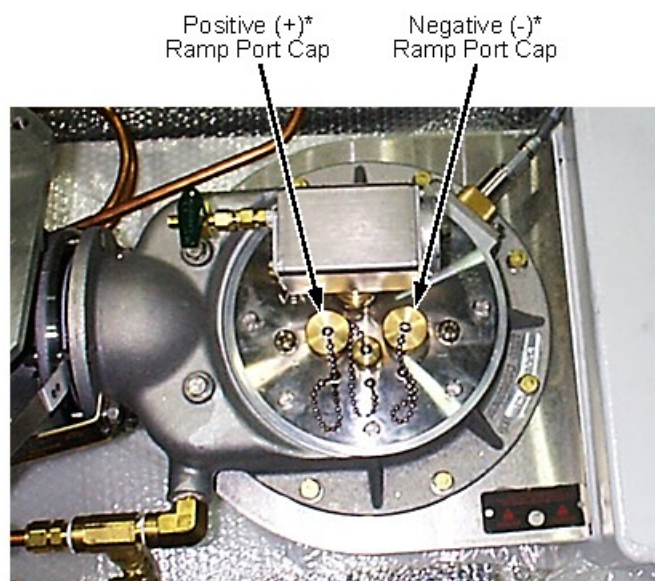
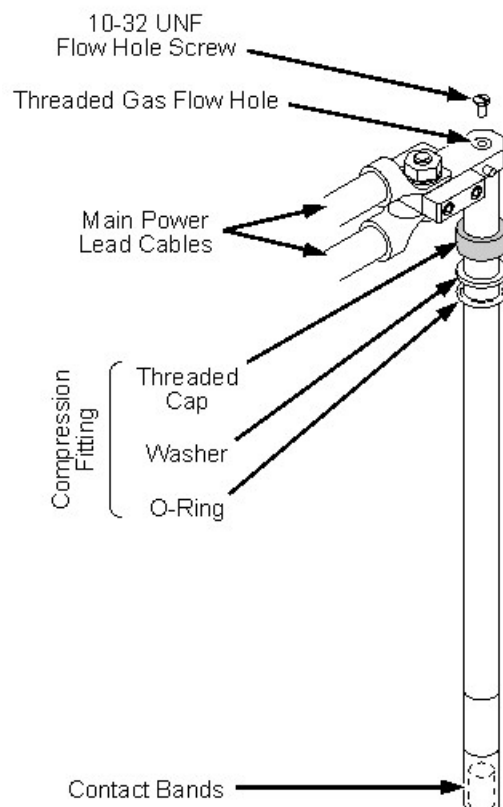


NOTICE

Replace the Ramp Port Caps immediately after removing Main Lead Extensions to prevent ice build-up inside the Vertical Penetration.

14. The steps below are performed **AFTER** the magnet is ramped down and **AFTER** AC power has been disconnected from the ramp power source.
 - a. Install a screw into the flow hole of **only one** of the Main Lead Extensions. See Illustration 5-21.

Illustration 5-21: Main Lead Extension and Ramp Ports



* Normal polarity shown.

- b. Remove all ice around the Ramp Lead Port compression fitting on the Main Lead Extension that is being removed (i.e. the Main Lead Extension that has the flow hole plugged in Step 14.a). See Illustration 5-21.

- c. Unscrew the Ramp Lead Port compression fitting and remove the Main Lead Extension from the magnet.
- d. Immediately replace the cap onto the Ramp Lead Port.
- e. Repeat Step 14.a through Step 14.d for the other Main Lead Extension.

6.4.2 Magnet Rampdown Using Auxiliary Cable

The Auxiliary Rampdown Cable (2217315) is used to rampdown a magnet when the Shim Lead cannot be engaged into the Shim Lead Connector (P100) in the magnet Vertical Penetration due to ice or damage to the P100's Sav-Con Connector. This cable connects the Main Coil Heater via the J1C connector on the magnet penetration and J3 on the Main Power Supply. Main Coil voltage will not be accessible with this wiring configuration. Use a digital voltmeter (DVM) with the Voltage Sense Leads connected to the Main Lead Extensions to approximate the Main Coil voltage.

6.4.2.1 Preparation for Rampdown (Auxiliary Cable)



NOTICE

Use the Auxiliary Rampdown Cable and procedure below ONLY when the Shim Leads cannot be engaged into the P100 connector.

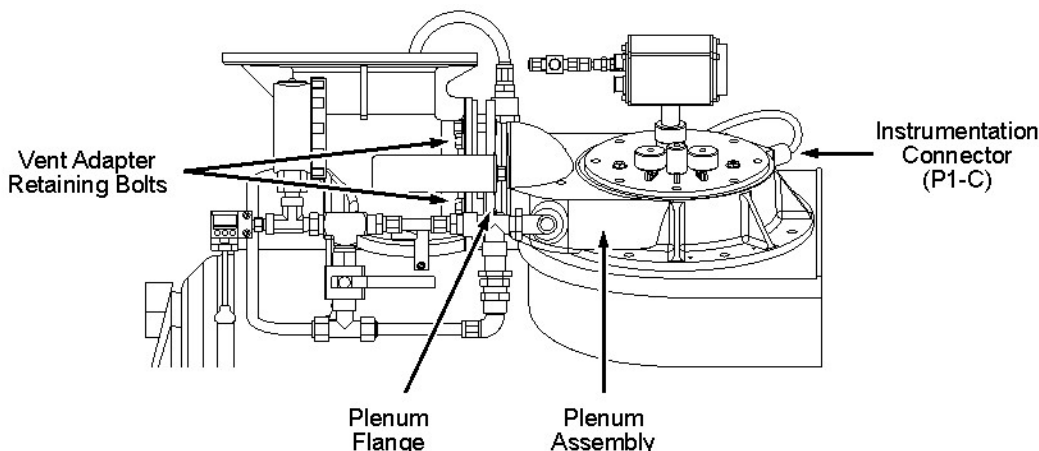


NOTICE

DO NOT contact the Shim Lead Connector (P100) or the Main Lead Pins with metallic or conductive tools or materials since the Auxiliary Rampdown Cable may leave current in the Superconductive Shim Coils. This current can transfer to the Main Coils.

1. "Top Fill" the magnet to $\geq 90\%$ helium level in conformance with Chapter 3, Section 3.2, Liquid Helium Fill, before continuing with this procedure.
2. Set up MetroLab Teslameter 46-255839P2 and #5 Teslameter Probe 46-255839P104 in conformance with Chapter 3, Section 4, Preparations for Field Measurement. The probe can be mounted to either a cardboard box or nonferromagnetic frame since only the overall field strength is monitored during ramping.
3. Perform the Ramp Power Supply checks in conformance with , Section , Main Power Supply Tests. Verify that the Shim Supply Heater currents are set at $810 \text{ mA} \pm 10 \text{ mA}$.
4. Make sure the input power cables for the power supplies are disconnected.
5. Replace the Main Lead Contact Bands in conformance to Section 4, Main Lead Extension Contact Band Replacement.
6. Disconnect the Instrumentation Connector (P1C) from connector J1C on the Plenum (Illustration 5-22). Connect Auxiliary Rampdown Cable 2217315 between connector J3 on the power supply (750 amp Main Power Supply: Illustration 5-17; 1000 amp Main Power Supply: Illustration 5-18) and connector J1C. See Chapter 6, Section 2.1, Magnet System Wiring.

Illustration 5-22: Instrumentation Connector (P1C) Location



6.4.2.2 Resistance Checks (Auxiliary Cable)



WARNING

POTENTIAL QUENCH HAZARD

THE MAGNET MAY QUENCH, RAPIDLY RELEASING CRYOGENIC GASES, IF ITS LIQUID HELIUM LEVEL DROPS TOO LOW DURING THIS PROCEDURE. MAKE SURE THE MAGNET $\geq 90\%$ FULL OF LIQUID HELIUM BEFORE STARTING THIS PROCEDURE.

1. Make sure the input power cable for the Magnet Power Supply is disconnected.
2. Connect the Main Power Supply and Main Lead Extensions to the magnet by making all cable connections in conformance with Chapter 3, Section 6, Connections for Ramping and Shimming, but using the Auxiliary Rampdown Cable instead of the Switch Heater Cable.
3. Set the Heater 1 Main and Heater 2 Shim Axial Switches to the "O" (off) position. See Illustration 5-17 (750 amp Main Power Supply) or Illustration 5-18 (1000 amp Main Power Supply).
4. Set the Current Adjust and Voltage Controls to "O" (full CCW).
5. Connect the input power cable to the Magnet Power Supply.



WARNING

POTENTIAL MAGNET QUENCH

TURNING ON THE MAIN HEATER SWITCH WHILE PERFORMING THESE RESISTANCE CHECKS WILL CAUSE A MAGNET QUENCH. MAKE SURE THE MAIN HEATER SWITCH STAYS OFF DURING RESISTANCE CHECKS.

6. Set the Main Power and Power On Switches to "ON."
7. Connect a digital voltmeter (DVM) to the end of the Voltage Sense Leads.

8. Set the Current Adjust Controls on the Magnet Power Supply to maximum (full CW).
9. Observe the Digital Current Meter and slowly turn the Voltage Control (CW) to set a 500 A current through the Main Leads, Main Lead Extensions and the persistent Main Switch.
10. Record the voltage reading on the DVM in Chapter 8, Section 5, Magnet Ramping and Parking Current Log.



CAUTION

Potential Magnet Quench

Voltage readings > 150 millivolts at 500 amps indicate unacceptable internal contact resistance at the Main Lead Extensions. Higher resistances will add more heat to the magnet, increasing boiloff and possibly causing a quench during ramping. Make sure the voltage readings are ≤ 150 mV at 500 amps.

11. If the DVM reads > 150mV, perform one or more of the following steps to reduce the contact resistance.
 - Wait approximately 1 minute with current running, readings may drop as the Main Lead Extensions cool.
 - Make sure the Hold-Down Tool is properly mounted and tightened.
 - Temporarily increase current to 700 amps, making sure that the voltage reading does not exceed 200 mV. Slowly decrease current back to 500 amps after reaching 700 amps or 200 mV.

If the reading still exceeds 150 mV:

1. Turn the Voltage and Current Adjust controls to zero (full CCW).
2. Turn off the Main Power Supply input power.
3. Tighten the bolts securing the Ramp Cables to the power supply and Main Leads Extensions.
4. Lift and reseal the Main Leads.

Repeatedly failing the contact resistance check may indicate one of the following:

- The Main Lead Extension Contact Bands replaced as directed in Chapter 3, Section 6, Connections for Ramping and Shimmiing, are not performing correctly and need replacing in conformance with Section 4, Main Lead Extension Contact Band Replacement.
- The Main Leads are damaged and need replacing.

If changes are made in Step 11, repeat Step 1 through Step 11.

12. Continue to Step 13 when the DVM voltage is ≤ 150 mV
13. Set the Voltmeter Select Switch to the "MAIN POWER SUPPLY" position and increase the current to 735 amps. If the voltage reading at 735 amps is > 2.2 volts:
 - a. Reduce the voltage/current to zero.

- b. Turn off the power supply off.
 - c. Remove, clean and tighten the ramp cable contacts at both the power supply and the Main Lead Extensions. Make sure the proper hardware is used to connect the ramp cables.
 - d. Repeat the resistance check in Step 11.
14. When the resistance check voltage reading in Step 13 is < 2.2 volts, turn the Voltage and Current Controls to zero and turn off the power supply.

6.4.2.3 Magnet Rampdown (Auxiliary Cable)



! WARNING

POTENTIAL QUENCH HAZARD
INCORRECT CONNECTION POLARITY AND POWER SUPPLY CURRENT SETTINGS CAN CAUSE THE MAGNET'S MAIN COILS TO QUENCH AND CAN BURN UP THE POWER SUPPLY WHEN TURNING ON THE MAIN SWITCH.
MAKE SURE THE CONNECTION POLARITY AND POWER SUPPLY CURRENT SETTINGS ARE THE SAME AS LAST RECORDED IN THE "MAGNET RAMPING AND PARKING CURRENT LOG".



! WARNING

POTENTIAL QUENCH HAZARD
IF THE AXIAL SHIM HEATER SWITCH IS NOT "ON" DURING THE ENTIRE MAIN FIELD ADJUSTMENT PROCESS, A MAGNET QUENCH CAN OCCUR AND IRREPARABLE DAMAGE SHIM COIL DAMAGE CAN BE DONE. THE POWER SUPPLY WILL NOT PASS CURRENT TO THE MAIN LEAD EXTENSIONS WITH THE AXIAL SHIM HEATER OFF.
MAKE SURE THE AXIAL SHIM HEATER SWITCH REMAINS "ON" DURING THIS ENTIRE PROCEDURE.



! WARNING

POTENTIAL QUENCH HAZARD
THE MAGNET MAY QUENCH IF THE MAIN POWER SUPPLY EXPERIENCES LARGE OUTPUT FLUCTUATIONS AND/OR EXCESSIVE RIPPLE.
MAKE SURE THE MAIN POWER SUPPLY IS ROUTINELY CALIBRATED AT AN APPROVED FACILITY.



NOTICE

Make sure the polarity recorded Chapter 8, Section 5, Magnet Ramping and Parking Current Log, matches the configuration of the Ramp Polarity Plate (Illustration 5-20). The "Normal" ramp polarity is "+" (positive) on the left and "-" (negative) on the right as viewed from the Coldhead side of the magnet. "Reversed" ramp polarity has "+" on the right and "-" on the left as viewed from the Coldhead side of the magnet. Call the National Support Center if the two do not match.



NOTICE

If a Quench occurs during change of magnetic field, immediately turn the Voltage Adjust and Current Adjust controls to zero.

- NOTE:** All main coil driving voltages provided in this procedure will be equal in magnitude but opposite in polarity for “reverse ramped” magnets. Normal polarity is shown in Illustration 5-16.
- NOTE:** Ice will build up around the Ramp Lead Hold-Down Tool flow holes during the rampdown procedure. Remove ice as needed to maintain helium gas flow through the flow holes.
1. Retrieve the Main Coil connection polarity and latest final parking current in the Chapter 8, Section 5, Magnet Ramping and Parking Current Log.
 2. Make sure the Main Leads are connected with the polarity indicated in Step 1.
- NOTE:** Main Coil voltage is not directly accessible with the Shim Lead connector disengaged. Use digital voltmeter (DVM) readings with the Voltage Sense Leads connected to the Main Lead Extensions for the steps below. This voltage approximates the Main Coil voltage and will prevent the rampdown voltage from exceeding the recommended value.
3. Make sure the DVM is connected to the Voltage Sense Leads (Illustration 5-21) from the Main Lead Extensions.
 4. Set the Current Adjust Controls on power supply to:
 - **for 750 amp power supply 46-260776G3:** maximum (full CW).
 - **for 1,000 amp power supply 46-260776G4:** 10 amps more than the parking current retrieved in Step 1.
- NOTE:** Power supply controls are shown in Illustration 5-17 (750 amp Main Power Supply) and Illustration 5-18 (1000 amp Main Power Supply).
5. Set the Voltage Control to adjust the power supply output current to the parking current value retrieved in Step 1 above.
 6. Set the Heater 1 Main Switch to “I” (on).
 7. Allow approximately 1 minute for the Main Switch to go normal.
 8. When the Main Switch is normal, **slowly** decrease the Voltage Control (CCW) until -0.3 V is observed on the DVM.
 9. Continue to adjust the power supply voltage using the Voltage Control during rampdown in conformance with the voltages read on the DVM and Main Coil current ranges shown in Table 5-5.

Table 5-5: Rampdown (Using Auxiliary Cable) Profile

Main Coil Voltage	Main Coil Current Range
-0.10 Volts	Start - 720 Amps
-0.30 Volts	720 - 700 Amps
-0.40 Volts	700 - 675 Amps
-0.50 Volts	675 - 650 Amps
-0.80 Volts	650 - 625 Amps
-1.40 Volts	625 - 500 Amps
Gradually turn the Voltage Control to zero (full CCW) then turn the Current Controls to zero (full CCW)	500 - 0 Amps

10. Observe the voltmeter reading with the power supply's Voltmeter Select Switch in the "MAIN COIL" position. A zero reading on the power supply's digital Voltage and Current Meters indicates that the magnet is fully discharged.
11. When the magnet is fully discharged, set the Heater 1 Main Switch to "O" (off).
12. Set the Main Power and Power On switches to "OFF."
13. Disconnect the input power cables from the magnet power supply.

NOTE: The Pressure Controller will maintain helium vessel pressure.



! WARNING

POTENTIAL ASPHYXIATION AND COLD BURN HAZARD!
IMPROPER ENGAGEMENT/DISENGAGEMENT OF THE MAIN LEAD EXTENSIONS WITH THE MAIN LEAD PINS CAN CAUSE A RAMPED MAGNET TO QUENCH. INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND, IF THE MAGNET IS AT FIELD, ALLOW THEM TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE THEY CONTACT THE MAIN LEAD PINS.



! WARNING

ELECTRICAL SHOCK HAZARD!
TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.



! WARNING

ELECTRICAL SHOCK HAZARD!
DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.

**WARNING**

POTENTIAL COLD BURN HAZARD
CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE.
WEAR NONABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION OR EXTRACTION OF THE MAIN LEAD EXTENSIONS.

**NOTICE**

Replace the Ramp Port Caps immediately after removing Main Lead Extensions to prevent ice build-up inside the Vertical Penetration.

14. The steps below are performed **AFTER** the magnet is ramped down and **AFTER** AC power has been disconnected from the ramp power source.
 - a. Install a screw into the flow hole of **only one** of the Main Lead Extensions. See Illustration 5-21.
 - b. Remove all ice around the Ramp Lead Port compression fitting on the Main Lead Extension that is being removed (i.e. the Main Lead Extension that has the flow hole plugged in Step 14.a). See Illustration 5-21.
 - c. Unscrew the Ramp Lead Port compression fitting and remove the Main Lead Extension from the magnet.
 - d. Immediately replace the cap onto the Ramp Lead Port.
 - e. Repeat Step 14.a through Step 14.d for the other Main Lead Extension.

6.5 Finalization

No finalization steps.

7 Cryostat Warm-Up

7.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

7.2 Overview

Only a few situations require warming up the magnet in the field, for instance removing an internal ice blockage or field-repairing a vacuum leak that requires re-evacuation. If a magnet warm-up is required, contact the Regional Field Service Engineer before proceeding.

NOTE: The Cryostat is designed to allow for the recovery of liquid helium (LHe); however, because of the nature of the failure modes requiring warm-up, the liquid helium is most likely boiled off. If the unit is less than 50% full of liquid helium, warm up the magnet without liquid helium recovery. If recovery of liquid helium is to be done, a prepared Dewar (cold and containing helium gas) is recommended for liquid helium recovery. A warm Dewar evaporates a significant quantity of liquid helium in the cooldown process.

7.3 Preliminary Requirements

7.3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Nonabsorbent protective clothing (long sleeve shirt and pants)	1 set per person	-	-	-
Nonferrous Shoes	1 pair per person	-	-	-
Copper Tubing (optional)	AR	-	-	-
Heat Gun (optional)	AR	-	-	-
Lakeshore Cryotronics 208 Digital Thermometer (46-301477G2), Lakeshore Cryotronics DRC-80 (46-265269G1), Low-Cost Shield Temperature Diode Box (46-317543G1) or equivalent temperature sensing device.	1	-	-	-
Standard tools may be used	AR	-	-	-
Warning Sign & Label Kit	1	-	46-258770G4	-
Portable Oxygen Monitor	1	-	2287000	-

Item	Qty	Effectivity	Part#	Manufacturer
Cryogen Safety Wear Kit	1	-	46-271137G1	-
Universal Fill Line Kit	1	-	46-294705G1	-
Liquid Helium Transfer Line Kit	1	-	46-294513G1	-
Helium Regulator and Hose Kit	1	-	46-306734G1	-
Nonmagnetic Helium Cylinder Cart	1	-	46-258150P1	-
Flexible Helium Hose, 40 foot (12.2 M)	1 or 2	-	46-271135P1	-
Universal Service Ladder/Platform	1	-	2319156	-

7.3.2 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Gaseous Helium, 235 SCF Nonferrous Cylinder	AR	-	-	-
Liquid Nitrogen Dewars (pressure-building preferred)	AR	-	-	-
Helium Dewars (prepared: cold and containing helium gas)	AR	-	-	-

7.3.3 Replacement Parts

No replacement parts required.

7.3.4 Safety



⚠ DANGER

POTENTIAL FATAL INJURY!

MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, BEFORE STARTING THIS PROCEDURE.



⚠ DANGER

POTENTIAL FATAL INJURY!

MAKE SURE TO FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, SECTION 2.4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS, BEFORE AND WHILE PERFORMING THIS PROCEDURE.



! DANGER

POTENTIAL FATAL INJURY!

HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET — SAFETY REQUIREMENTS, SECTION 2.5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION, BEFORE STARTING THIS PROCEDURE.



! DANGER

FATAL EXPLOSIVE HAZARD!

GAS CYLINDER RUPTURE/DAMAGE OR OPENING CYLINDER VALVE FAST WILL RELEASE 2400 PSI GAS FROM CYLINDER. MAKE SURE GAS CYLINDERS ARE SECURE FROM FALLS AND AWAY FROM MOVING OBJECTS BEFORE USING. ALWAYS OPEN CYLINDER VALVE SLOWLY.



! WARNING

ASPHYXIATION & COLD BURN HAZARD!

A MAGNET QUENCH DURING PROCEDURE COULD RESULT IN RAPID EXPULSION OF LIQUID HELIUM THROUGH THE VERTICAL PENETRATION. MAKE SURE MAGNET IS RAMPED DOWN TO ZERO FIELD BEFORE STARTING THIS PROCEDURE.



! WARNING

ASPHYXIATION HAZARD!

PROCEDURE GENERATES ODORLESS, COLORLESS HELIUM GAS THAT DISPLACES OXYGEN IN THE AIR.

MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS ON — OR THE HATCH IS OPENED IF A MOBILE VAN — AND THAT THE MAGNET ROOM DOORS ARE SECURED IN THE OPEN POSITION BEFORE WARMING UP THE MAGNET.



! WARNING

POTENTIAL COLD BURN OR ASPHYXIATION HAZARD

INTERNAL CRYOSTAT PRESSURE BUILD-UP TO > 5.25 PSIG WILL OPEN 5.25 PSIG RELIEF VALVE.

BEFORE REMOVING EITHER RAMP OR FILL PORT CAPS OR LOOSENING ANY COMPONENT RESULTING IN THE RELEASE OF CRYOGEN PRESSURE 1) OBSERVE THE INTERNAL CRYOSTAT PRESSURE GAUGE READING, 2) SET MAGNET MONITOR PRESSURE TO 0.5 PSIG AND 3) VENT CRYOSTAT DOWN TO 0.3 PSIG THROUGH VENT VALVE V2.



! WARNING

POTENTIAL COLD BURN HAZARD

EXPOSURE TO CRYOGENS AND CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE.

WEAR PROTECTIVE CLOTHING, NONABSORBENT GLOVES AND GOGGLES WHEN PERFORMING THIS PROCEDURE.



NOTICE

Do not warm up the magnet if it is to be shipped from the site!

Make sure the MAC Team Leader is present and has determined that magnet warm-up is required before proceeding.

Make sure all required equipment is on site. Enough prepared Dewars to handle liquid helium recovery must be available if liquid helium (LHe) is to be recovered.

NOTE: If the magnet has Magnet Monitor link, turn the vessel pressure off in the 'Configuration/Alarms' tab.

7.3.5 Required Conditions

Condition	Reference	Effectivity
Make sure the area is secure and that ALL required Warning Signs are posted before starting this procedure.	-	-

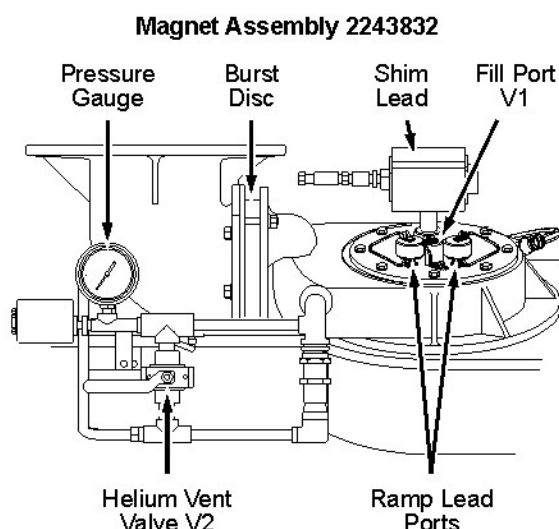
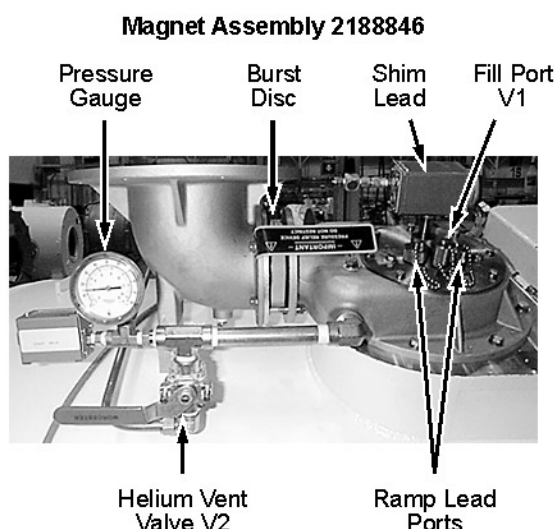
7.4 Procedure

7.4.1 Cryostat Warm-Up without Liquid Helium Recovery

NOTE: The following procedure must be used when the magnet's liquid helium level is < 50%.

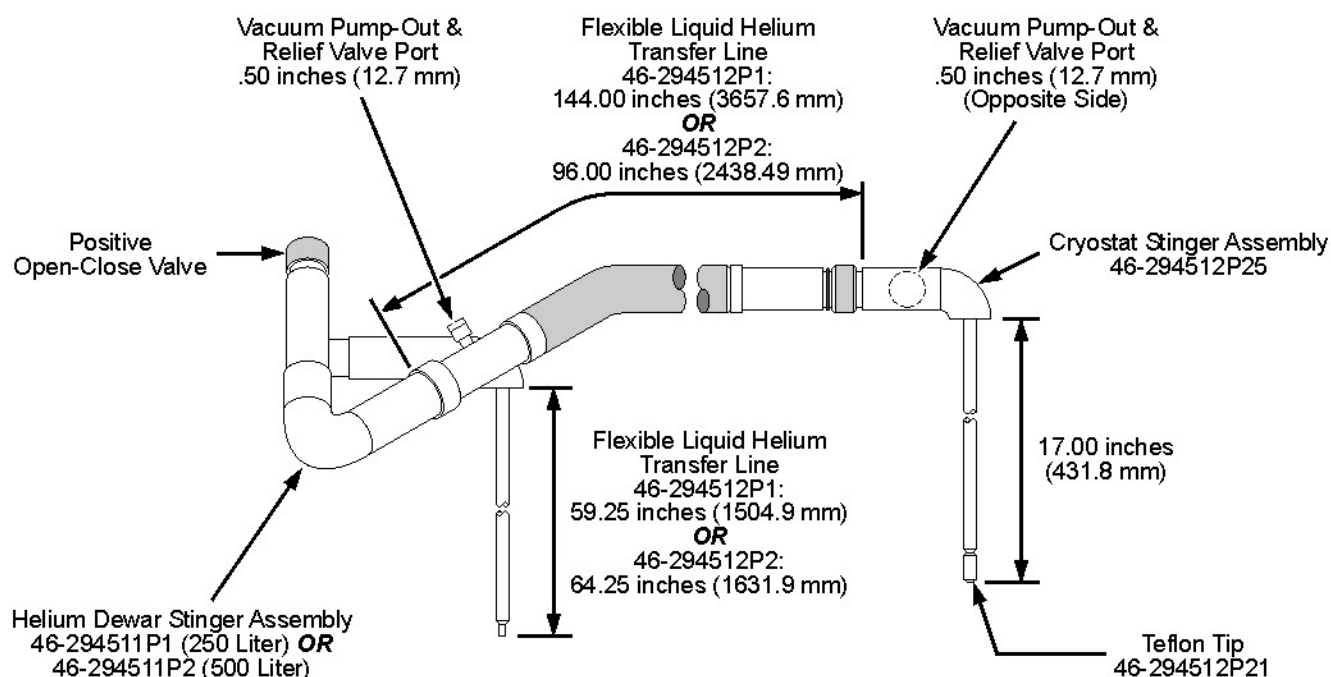
1. Turn off the Cryocooler Compressor and disconnect the power cable. Disconnect the supply and return gas lines at the Coldhead.
2. Open helium Vent Valve V2. See Illustration 5-23.

Illustration 5-23: Helium Vent Plumbing



3. Remove the Burst Disc in conformance with Section 11.1, Burst Disc (2349529) Replacement (or Section 11.2, Burst Disc (2194545-7) Replacement, where appropriate), and then reconnect the Vent Adapter.
4. Connect the Lakeshore Cryotronics DRC-80 46-265269G1), the Lakeshore Cryotronics Model 208 Thermometer Kit (46-301477G1) or the Low-Cost Shield Temperature Diode Box (46-317543G1) to monitor silicon diode temperature in conformance with Chapter 3, Section 2, Magnet Commissioning Checks.
5. Connect a helium gas (99.995%) and regulator set-up to the Helium Fill Port at Valve V1. See Illustration 5-23.
6. Insert a 7 inch (178 mm) Cryostat Stinger (46-294512P3) with a 16 inch (406 mm) Cryostat Stinger Extension (46-294512P12) into the Fill Port, and then tighten the compression fitting. See Illustration 5-24.

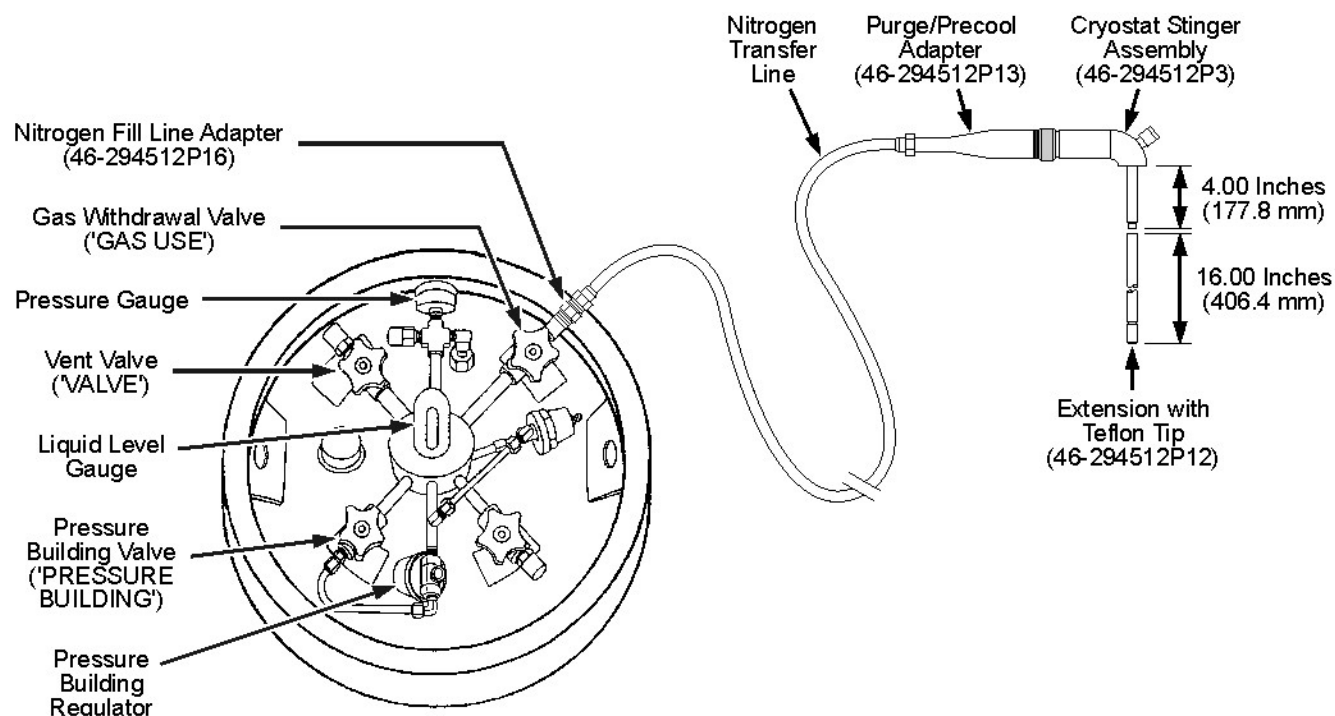
Illustration 5-24: Helium 'Bottom Fill' Equipment



7. Blow warm helium gas regulated at 4 - 6 psig through the Cryostat until the silicon diode temperature readout exceeds 90 K.

NOTE: The gas may be warmed through the heat exchange coil heated with a heat gun.

8. Shut off the helium gas flow using the Positive On/Off Valve. See Illustration 5-24.
9. Remove the helium gas cylinder & regulator set-up and connect the nitrogen gas set-up (Illustration 5-25) in its place.

Illustration 5-25: Nitrogen Gas Set-Up

10. Open the Positive On/Off Valve.
11. Start a nitrogen gas flow regulated at 4 - 6 psig and continue until the silicon diode temperature readout exceeds 290K.
12. Discontinue the nitrogen gas flow, remove the Stinger and cap Helium Fill Port V1. Close V2.

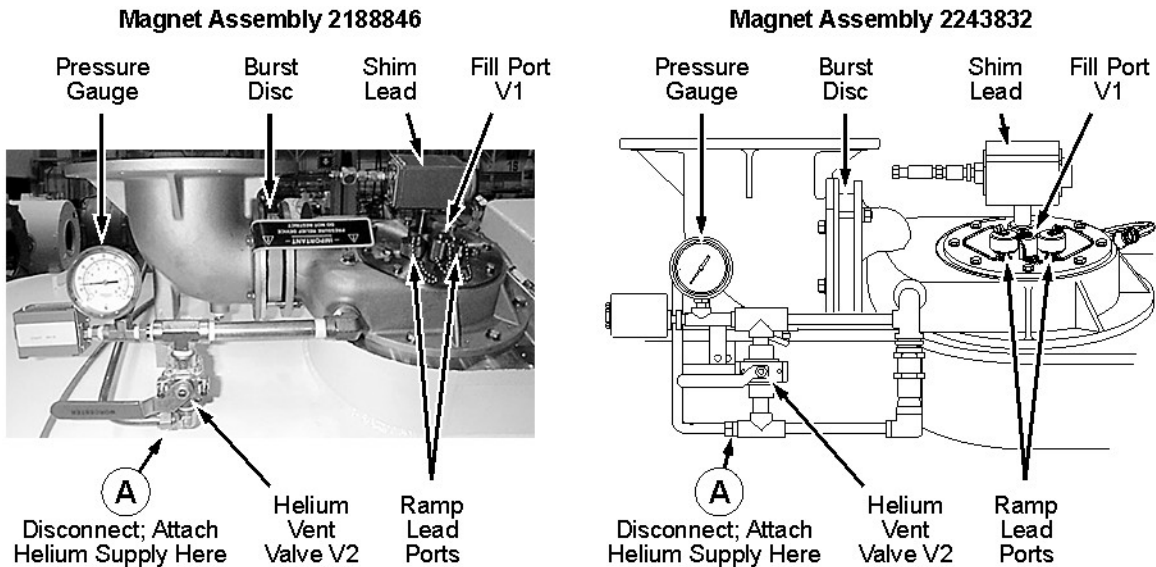
7.4.2 Cryostat Warm-Up with Liquid Helium Recovery**NOTICE**

Do not use this procedure if the magnet's liquid helium level is < 50%.

1. Check the magnet's liquid helium level. If the level is < 50%, use the warm-up procedure in Section 7.4.1.
2. Turn off the Cryocooler Compressor and disconnect the power cable. Disconnect the supply and return gas lines at the Coldhead.
3. Connect the Lakeshore Cryotronics DRC-80 (46-265269G1), the Lakeshore Cryotronics Model 208 Thermometer Kit (46-301477G1) or the Low-Cost Shield Temperature Diode Box (46-317543G1) to monitor silicon diode temperature in conformance with Chapter 3, Section 2, Magnet Commissioning Checks.
4. Open Vent Valve V2 momentarily to purge the vent plumbing and reduce internal pressure to < 0.3 psig. Close V2. See Illustration 5-23.

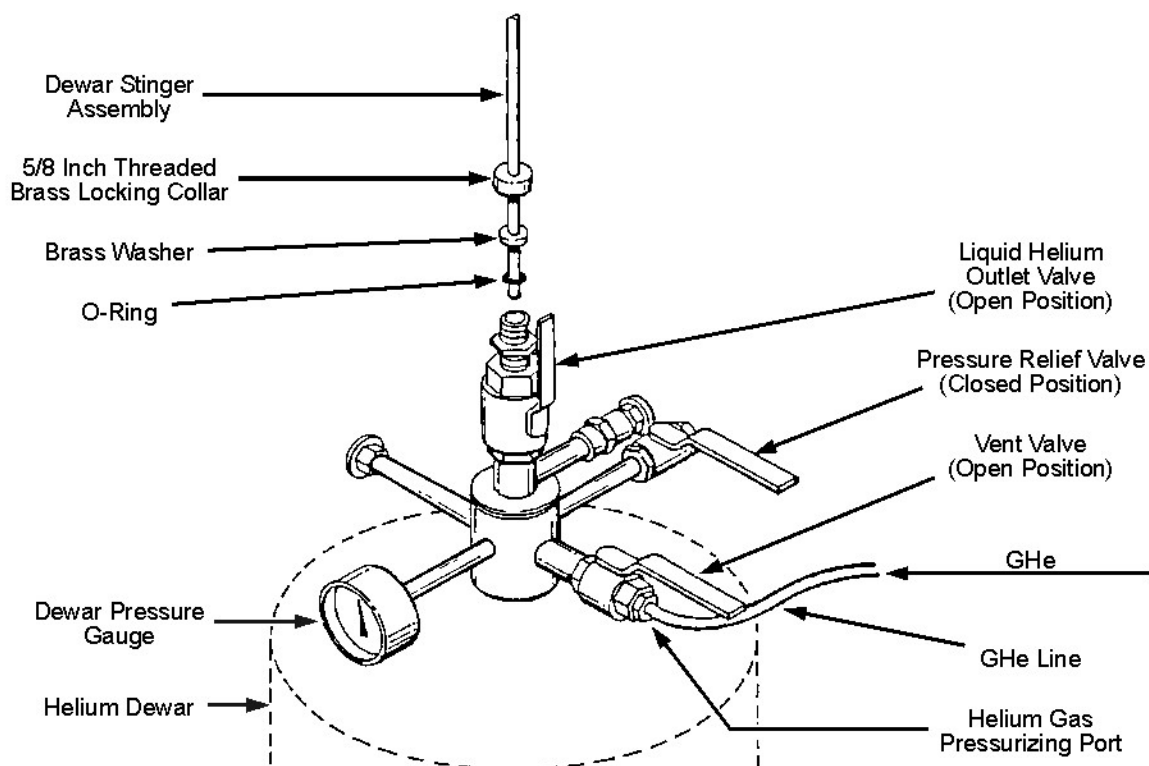
5. Disconnect the helium vent plumbing at "A" in Illustration 5-26 and connect the helium gas (99.995%) and regulator set-up at that point.

Illustration 5-26: Location for Helium Gas Connection



6. Loosen and remove the cap from Helium Fill Port V1. See Illustration 5-26.
7. Insert a 7 inch (178 mm) Cryostat Stinger (46-294512P3) with a 16 inch (406 mm) Cryostat Stinger Extension (46-294512P12) into the Fill Port, and then tighten the compression fitting. See Illustration 5-24.
8. Place the locking collar, washer and o-ring from a prepared helium Dewar onto the other end of the transfer line. See Illustration 5-27.

Illustration 5-27: Helium Transfer Line Extension Connection



9. Open Valve V2.

10. Blow helium gas regulated at 4 - 6 psig through the setup and into the Cryostat.

11. Set the valves on the Dewar to those shown in Illustration 5-27:

- Helium Outlet Valve - Closed
- Pressure Relief Valve - Closed
- Helium Vent Valve - Open

12. Wait for a bluish plume to start expelling from the transfer line, then:

- a. Open the Helium Outlet Valve.
- b. Fully insert the transfer line into the Dewar.
- c. Back off the transfer line approximately 1 inch (25 mm) from the bottom of the Dewar.
- d. Engage and lock the locking collar, washer and o-ring, securing the Transfer Line onto the Dewar.

NOTE: If a warm Dewar is used, insert and lock the transfer line onto the Dewar immediately for maximum cooling and flushing impact.

13. Monitor the cryogen level, gas pressure and Vent Valve to establish when the Dewar has filled and when the Cryostat is empty of liquid helium.

14. When the Dewar is full or the Cryostat is empty, turn off the gas flow to the Cryostat and close Valve V2.
15. Close the Positive On/Off Valve on the transfer tube.
16. Loosen the Dewar's compression fitting and withdraw the transfer line from the Dewar.
17. Set and wire the valves on the Dewar in the following positions:
 - Helium Outlet Valve - Closed
 - Pressure Relief Valve - Open
 - Helium Vent Valve - Closed
18. For continued liquid helium recovery with another prepared Dewar, repeat Step 9 through Step 17.

7.5 Finalization

No finalization steps.